



AIRBNB LINEAR REGRESSION

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01

CASE BACKGROUND

Overview of Airbnb and
the Case Study

Airbnb Background



Rise of Airbnb

Founded in 2008, Airbnb was a San Francisco-based online platform that connected property owners with travelers seeking a home-like stay away from home.

Property owners wanted to list their properties so they could invite travelers and charge them a premium to treat the space as their own.

The goal of Airbnb's aspiring hosts was to use the Airbnb website to attract these travelers who were willing to pay high premiums for the comfort of staying in a home as opposed to a hotel.

Airbnb's Business Model



Instead of Airbnb owning the properties directly, Airbnb used a two-sided online marketplace business model that connected hosts with guests through Airbnb's website.

Airbnb facilitates transactions between the host and the guest, charging service fees for both users on the website to generate revenue. Guests typically pay a 6-12% fee for booking, while hosts pay around 3%.

These are the main factors the business model relies on:

- **All Listing Located on Airbnb:**
 - This allows Airbnb to collect large amounts of text data for the marketing team to use to stay ahead of competitors.
- **Fast Scalability:**
 - Because Airbnb does not own the properties, it can expand globally without worrying about overhead.
- **Guest Satisfaction Reviews:**
 - Guest reviews and ratings help build both credibility and demand for hosts and location preferences.

Case Study

Business Problem:

Can basic ratings from guest review text data provide meaningful insight to improve host performance and increase profits?

Goal:

To determine if Airbnb can use customer review text data to improve performance on the website and increase net profits.

The case study focuses on whether analyzing the review sentiment through text mining can help Airbnb and its hosts make better pricing, listing, and location-specific strategies for consumers.





02

LINEAR REGRESSION MODEL

Methodology and Building
the Linear Regression Model

Data Overview

MIAMI	
n	2,000 listings
Avg. occupancy	35.25 days / 90
Avg. price	\$172.51 / night
Avg. rating	4.65
Superhost	42% of listings
Entire home	77% of listings
Sentiment mean	7.10

PARIS	
n	2,000 listings
Avg. occupancy	76.10 days / 90
Avg. price	\$101.04 / night
Avg. rating	4.58
Superhost	15% of listings
Entire home	86% of listings
Sentiment mean	9.69

Variable Construction & Transformations

01

Dependent variable

$\ln(\text{occupancy} + 1)$

Occupancy is right-skewed with 248 zeros in Miami. Log-transforming normalizes the distribution. The +1 prevents $\log(0)$.

02

Continuous predictors

$\ln(\text{price})$, $\ln(\text{reviews} + 1)$, $\ln(\text{accommodates})$, $\ln(\text{min.nights} + 1)$

Log-transformed because price and review counts are heavily right-skewed.

Coefficients become elasticities: a 1% increase in $X \rightarrow \beta\%$ change in occupancy.

03

Binary (dummy) predictors

Superhost, Pro host, Entire home, Instant bookable

Kept as 0./1 indicators. Coefficient = approximate % change in $\ln(\text{occupancy})$ when switching from 0 to 1.

Miami – Correlation matrix

Size variables (accommodates, bedrooms, bathrooms, beds) were checked for correlation.

	accommodates	bedrooms	bathrooms	beds
accommodates	1.00	0.87	0.75	0.85
bedrooms	0.87	1.00	0.82	0.83
bathrooms	0.75	0.82	1.00	0.71
beds	0.85	0.83	0.71	1.00

Miami shows very high correlations among all four size variables ($r = 0.71$ to 0.87). To avoid multicollinearity, only $\ln(\text{Accommodates})$ is included in the Miami regression. Bedrooms, bathrooms, and beds are excluded.

Paris - Correlation matrix

Size variables (accommodates, bedrooms, bathrooms, beds) were checked for correlation.

	accommodates	bedrooms	bathrooms	beds
accommodates	1.00	0.01	0.00	-0.02
bedrooms	0.01	1.00	0.52	0.67
bathrooms	0.00	0.52	1.00	0.63
beds	-0.02	0.67	0.63	1.00

In Paris, Accommodates is nearly uncorrelated with the other three variables ($r \approx 0.00$). Bedrooms, bathrooms, and beds are correlated with each other ($r = 0.52\text{--}0.67$) but not with Accommodates. The Paris regression therefore includes both $\ln(\text{Accommodates})$ and Bedrooms as separate predictors.

Constructing the Sentiment Variable

01

Load reviews

Import guest review text (comments) from Miami_reviews.csv and Paris_reviews.csv

02

Text processing

Strip whitespace → lowercase → remove stop words → stem words to root forms

03

Score with syuzhet

get_sentiment(method="syuzhet") assigns each review a score from negative or positive by summing dictionary word values

04

Assign to listing

One score per listing, appended directly to the listings dataframe as listings\$sentiment

Why Sentiment?

Star ratings compress guest experience into a single number. Sentiment captures the actual language guests use - 'breathtaking views', 'spotless', 'noisy neighbours' - encoding richer quality signals beyond what a 4.5 stars can convey.

7.10

Miami mean
std: 3.86

9.69

Paris mean
std: 6.50

Variable selection

Variable	Miami	Paris
ln(Price)	Included	Included
ln(Number of reviews +1)	Included	Included
Rating	Included	Included
ln(Accommodates)	Included	Included
Bedrooms	Excluded - collinear with accommodates (r = 0.87)	Excluded - not significant (p = 0.71)
ln(Min. nights +1)	Included	Included
Superhost	Included	Excluded - not significant (p = 0.24)
Pro host	Included	Included
Entire home	Included	Included
Instant bookable	Included	Included
Sentiment	Included	Included

Overall Model Fit

MIAMI	
R^2	0.1792
Adjusted R^2	0.1751
F-statistic	43.44
p-value	< 2.2e-16
Predictors	10
n	2,000

PARIS	
R^2	0.1743
Adjusted R^2	0.1705
F-statistic	46.67
p-value	< 2.2e-16
Predictors	9
n	2,000

Miami – Regression Coefficients

Variable	Coef.	Std. Error	p-value	Sig.
Intercept	0.817	0.344	0.018	*
ln(Number of reviews +1)	0.089	0.033	0.008	**
ln(Price)	-0.190	0.059	0.001	**
Rating	0.218	0.050	<0.001	***
ln(Accommodates)	0.462	0.077	<0.001	***
ln(Min. nights +1)	0.089	0.040	0.026	*
Superhost	0.543	0.067	<0.001	***
Pro host	-0.156	0.071	0.029	*
Entire home	0.642	0.087	<0.001	***
Instant bookable	0.348	0.069	<0.001	***
Sentiment	0.028	0.009	0.001	**

Paris – Regression Coefficients

Variable	Coef.	Std. Error	p-value	Sig.
Intercept	5.483	0.144	<0.001	***
ln(Price)	-0.385	0.032	<0.001	***
ln(Number of reviews +1)	-0.071	0.013	<0.001	***
Rating	0.082	0.021	<0.001	***
ln(Accommodates)	0.153	0.039	<0.001	***
ln(Min. nights +1)	-0.105	0.021	<0.001	***
Pro host	-0.329	0.046	<0.001	***
Entire home	0.253	0.043	<0.001	***
Instant bookable	0.070	0.032	0.029	*
Sentiment	0.006	0.003	0.026	*

Key Variables Interpretations

	Miami: coef: -0.190 (p = 0.001) Paris: coef: -0.385 (p < 0.001)	Price reduces occupancy in both markets, but the effect is two times stronger in Paris. A 10% price increase is associated with a -1.9% occupancy drop in Miami vs. -3.9% in Paris. Paris renters are far more price-sensitive, likely because of abundant alternatives (hotels, other Airbnbs). In Miami, strong vacation demand absorbs price increases more easily.	Price One of the most striking market differences. In Miami, the Superhost badge is among the strongest predictors - it signals trustworthiness to leisure travelers choosing between unfamiliar properties. In Paris, it has no measurable effect. Paris renters appear to rely on other signals (price, reviews) rather than Aurbnb's certification.
	Entire home	Miami: coef: +0.642 (p < 0.001) Paris: coef: +0.253 (p < 0.001)	Full-property listings outperforms shared spaces in both cities - the preference for privacy is universal. The effect is 2.5 times larger in Miami (0.642 vs. 0.253), reflecting Miami's vacation culture where exclusive use of a property is a core expectation. In Paris, entire homes already dominate at 86% listings, reducing the contrast.

Sentiment – Results & Interpretation

MIAMI	
0.028 coefficient	p = 0.001 ** Std. error: 0.009 t = 3.234
A one-unit increase in sentiment score is associated with a +2.8% increase in ln(occupancy+1). Listings with more enthusiastic, positive review language attract more bookings over and above what the star rating captures - especially important in Miami where guests rely heavily on peer cues.	

PARIS	
0.006 coefficient	p = 0.026 * Std. error: 0.003 t = 2.228
Sentiment is significant in Paris but with a smaller coefficient (0.006 vs. 0.028). The Paris market shows wider sentiment variance, including some genuinely negative reviews. Review tone carries a real, independent signal - even after controlling for price, ratings, and host characteristics.	

Key insight

Sentiment is significant in both markets, confirming that the qualitative tone of reviews encodes information about occupancy that star ratings alone do not fully capture. The effect is stronger in Miami, consistent with that market’s greater reliance on peer signals and trust cues.

Were the results expected?



As expected

Price lowers occupancy, rating increases occupancy, entire home increases occupancy, larger properties increase occupancy - all directionally predicted and confirmed in both markets.



As expected

Instant bookable increases occupancy in Miami: removing booking friction should increase conversions especially for impulsive leisure travelers.



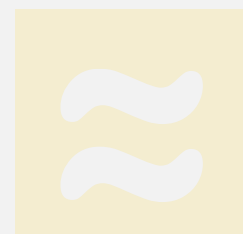
As expected

Superhost increases occupancy in Miami: trust signals should matter more for vacation markets where guests are unfamiliar with the area and properties.



As expected

Sentiment increases occupancy: properties with more positive review language should attract more guests over and above the numerical rating.



Partially expected

$R^2 = 17\text{-}18\%$: Models explain a meaningful but modest share of variance. Expected given the large role of unobservable (photos, neighbourhood, seasonality).

Surprising findings

Reviews lower occupancy in Paris (coef: -0.071 , $p < 0.001$)

Surprisingly, more reviews are associated with lower occupancy in Paris. The likely explanation: heavily-reviewed listings tend to be older, saturated properties that have lost freshness relative to newer competitors. In a market with very high baseline occupancy (76%), newer listings capture demand at the expense of established ones. Miami shows the opposite ($+0.089$), where reviews serve as a social proof for a more uncertain leisure market.

Min. nights increases occupancy in Miami (coef: $+0.089$, $p = 0.026$)

Higher minimum stay requirements should hurt bookings by restricting eligible guests – yet in Miami, the relationship is positive. Properties with higher minimum cater to week-long vacation stays, which dominate demand in Miami's leisure market. The minimum stay requirement is a proxy for a vacation-rental positioning that attracts committed, high-intent guests rather than deterring them.

Pro host lowers occupancy in both markets

Professional hosts manage multiple properties commercially, yet they perform worse than individual hosts on occupancy (Miami: -0.156 ; Paris: -0.329). Guests appear to value the authenticity and personal touch of owner-hosted listings. In Paris especially, the strong negative effect (-0.329) suggests that professionalization actively undermines the experience renters are seeking. The core Airbnb value proposition – 'live like a local' – may be eroded by commercial management.



03

TOPIC MODEL ON MIAMI AND PARIS REVIEWS

Topic Identification and
Cross-Market Comparison

Methodology



01 Text Preprocessing

Remove custom stopwords

Airbnb · Paris · Miami

Lowercase → Tokenize

Remove English stopwords

Stem words (Porter)

1,819 Miami reviews

1,813 Paris reviews



02 STM Model (K=0)

R package: stm

Topic selection: K=0

Algorithm auto-detects optimal K

Spectral initialization

Converged ~68 iterations

→ 81 topics (Miami)

→ 94 topics (Paris)



03 Topic–Rating Linkage

estimateEffect() in R

Models each topic's association
with star ratings

5-star vs. 3-star reviews

Output: which topics appear
more in high/low reviews

MARKET DIVERSITY: Topic & Vocabulary Scale Comparison

- Paris generates 16% more topics with 53% richer vocabulary from roughly the same number of reviews.
- Paris: international guests, multilingual reviews, diverse traveler types (tourists, business, cultural visitors).
- Miami: reviews are more homogeneous, predominantly English-language, leisure/vacation framing, beach/relaxation themes

	Miami	Paris
Topics	81	94
Documents	1,819	1,813
Unique word	3,060	4,696
Total tokens	102,754	184,893



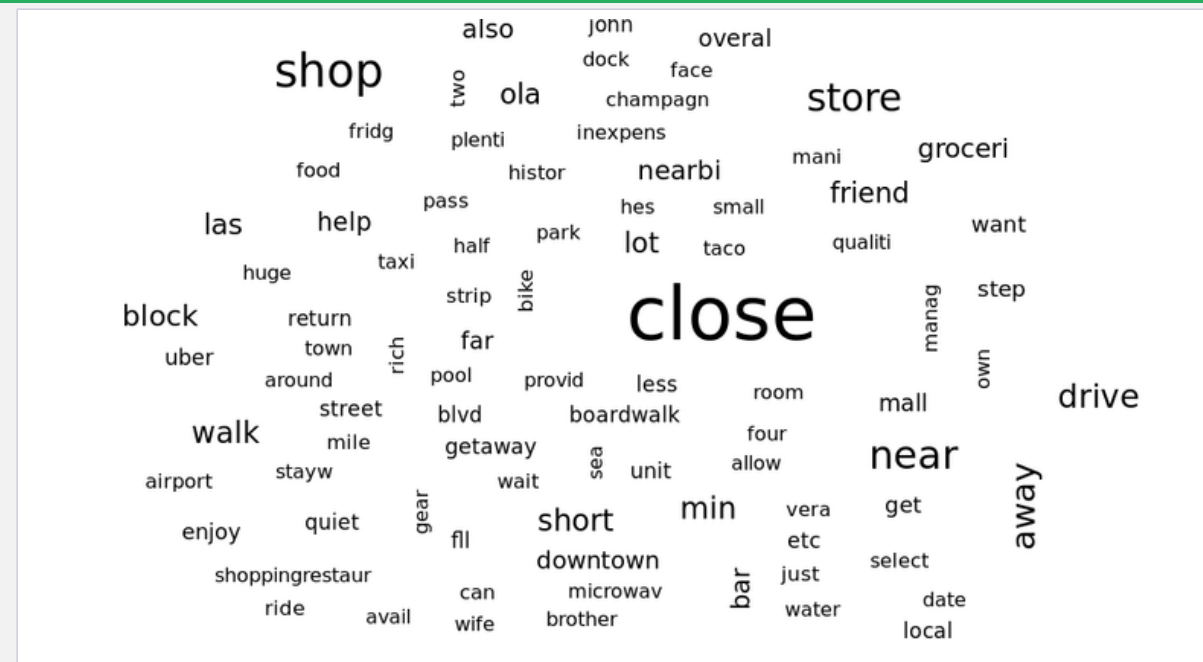
Miami: Topics Linked to Higher & Lower Ratings

▲ POSITIVE TOPICS — Associated with 5-Star Reviews



T55 +0.025

Everything As Advertised



T19 +0.015

Prime Beach Location & Walkability



T10 +0.014

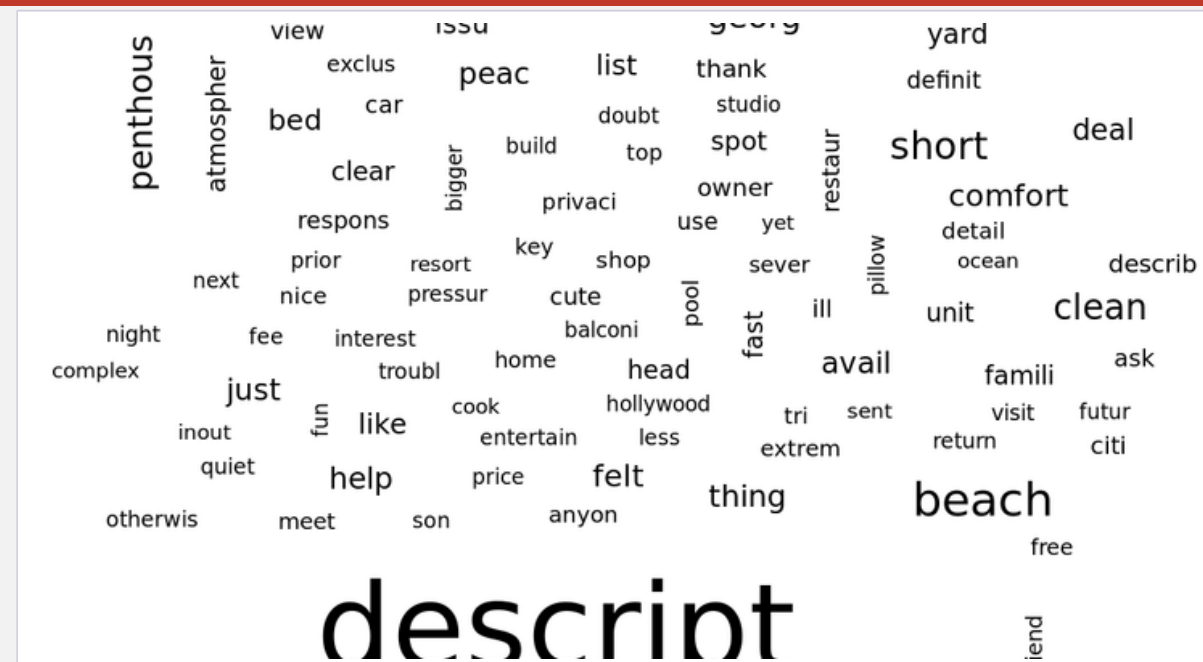
Exceptional Host Warmth

▼ NEGATIVE TOPICS — Associated with 3-Star Reviews



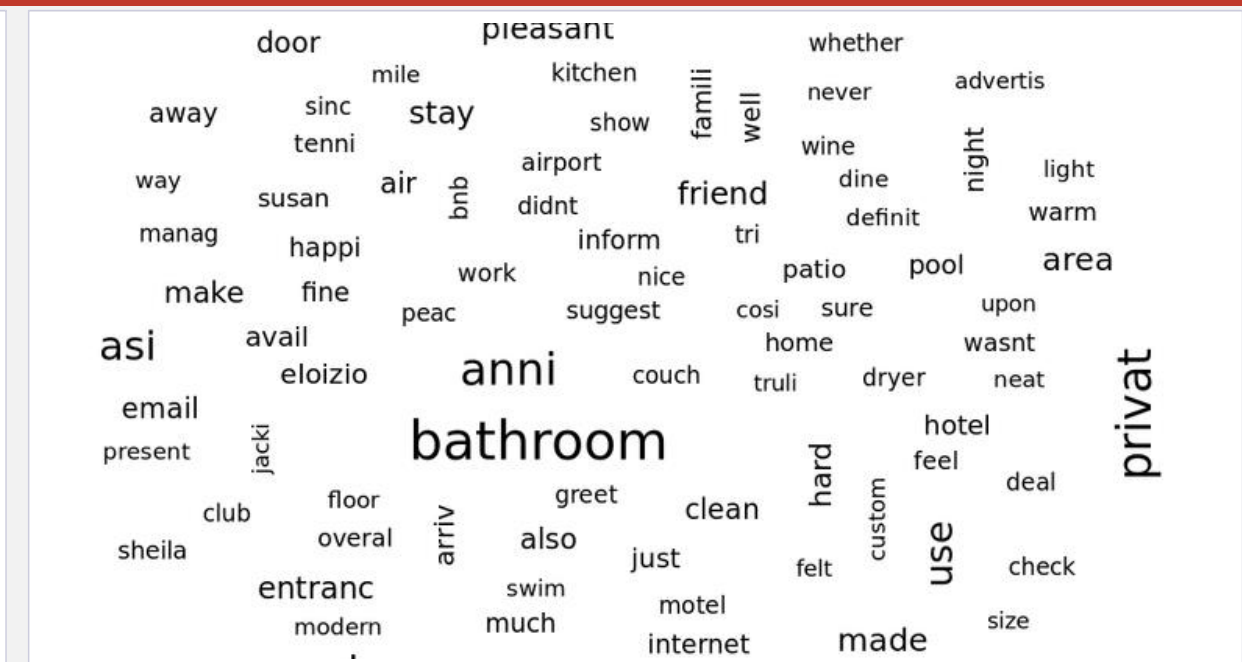
T28 -0.143

Property Not Delivering



T38 -0.080

Listing Inaccuracy



T4

Private Room Disappointment

⚠️ Miami Pattern: MARKETING FAILURES — guests let down by overpromising listings and unmet property expectations

Paris: Topics Linked to Higher & Lower Ratings

▲ **POSITIVE TOPICS** — Associated with 5-Star Reviews



T90 +0.022

Outstanding Overall & Recommendation



T10 +0.018

Stylish Parisian Apt -- Le Marais



T87 +0.014

Warm Host & Home-Like Feeling

▼ NEGATIVE TOPICS — Associated with 3-Star Reviews



T85 -0.028

No Elevator / Walk-Up Stairs



T39 -0.027

Small Space & Street Noise

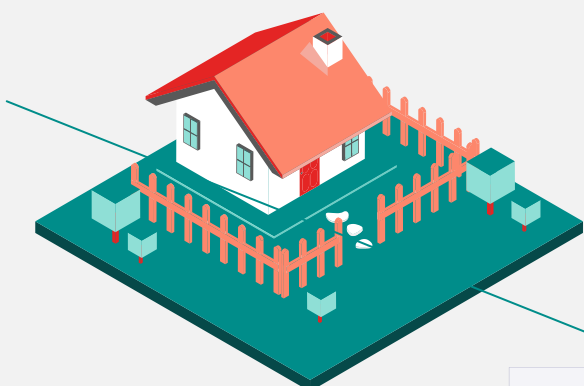


T37 -0.026

Cleanliness & Bathroom Failures

⚠️ Paris Pattern: PHYSICAL CONSTRAINTS — building limitations, noise & hygiene guests discover on arrival

Cross-Market Topic Comparison



Miami

	 MIAMI	 PARIS
#1 Positive Driver	Listing accuracy - meets expectations	Overall enthusiasm & strong recommendation
#2 Positive Driver	Prime beach location & walkability	Stylish Parisian apt - Le Marais identity
Universal Positive ✓	Host Warmth	Host Warmth
#1 Negative Driver	Property not delivering (pool, maintenance)	No elevator — walk-up stairs (3+ floors)
#2 Negative Driver	Listing inaccuracy - misleading ads	Small space & street noise
Root Cause	⚠️ MARKETING FAILURES	⚠️ PHYSICAL CONSTRAINTS

The Universal Truth: Host Warmth

Regardless of market, culture, or property type — a warm, welcoming host is the single most consistent driver of 5-star reviews.



Paris

04

RECOMMENDATIONS AND RESEARCH OPPORTUNITIES

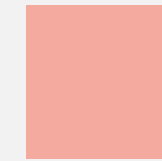
Marketing Strategy
Recommendations and
Future Marketing Research
Directions

Miami Marketing Recommendations



Grow Superhost Adoption

Superhost is one of the strongest drivers of occupancy in Miami, so Airbnb should help more hosts qualify and maintain status



Enforce Listing Accuracy

Miami's main low-rating themes were expectation mismatch and listings failing to deliver as promised



Expand Instant Booking Usage

Instant-bookable listings perform better, so Airbnb should reduce booking friction wherever possible



Promote Personal Hosting

Professional hosts underperform, so Airbnb should highlight local, personal hosts as a point of differentiation

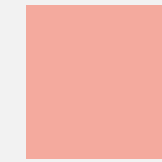
Miami Strategy = build trust, reduce friction, and stop over promising

Paris Marketing Recommendations



Require Better Physical Disclosure

Paris's main low-rating topics were no elevator, small space, and street noise, so guest need clearer expectations before booking



Support Competitive Pricing

Paris guests are more price-sensitive, so pricing tools and benchmarking matter more here



Improve Cleanliness Enforcement

Cleanliness and bathroom failures were a major negative theme and can be fixed directly by hosts



Protect the Personal Host Experience

Professional hosting performs worse, so Airbnb should position owner-hosted stays as part of the Paris experience

Paris strategy = focus on transparency, pricing, and guest expectations

Gaps & Limitations of the Current Analysis

Limitations	Why It Matters
One-time 90-day snapshot	Makes it difficult to separate seasonal effects from long-term market behavior
Correlation does NOT equal causation	The model identifies relationships, but not direct causal effects
No visual or photo data	Listing photos likely influence bookings, but were not included in the analysis
Review age not accounted for	Older reviews may not reflect current listing quality

While the regression and topic modeling identified meaningful patterns, several important drivers of Airbnb performance remain unmeasured, creating opportunities for future research.

Recommended Future Research Projects

Listing Quality	Host Scaling	Pricing & Seasonality	Listing Age
Purpose: Analyze listing photos and presentation quality Value: Helps reduce mismatch in guest expectations	Purpose: Analyze how scaling affects host quality Value: Improve consistency across listings	Purpose: Analyze seasonal booking trends Value: Support better pricing strategies	Purpose: Understand how older listings perform over time Value: Improve older listing performance
ROI: HIGH Miami mismatch topic: -0.143 Listing inaccuracy was one of the strongest negative drivers of guest reviews	ROI: MEDIUM-HIGH Paris: -0.328 Miami: -0.156 Professional hosts were associated with lower occupancy in both markets.	ROI: HIGH Paris: -0.391 Miami: -0.190 Paris showed stronger price sensitivity, highlighting the importance of pricing strategy.	ROI: MEDIUM Paris: -0.075 Miami: $+0.089$ Review effects reversed between Miami and Paris.

Pricing & Seasonality and Listing Quality stand out as the highest ROI opportunities because both can be addressed immediately and directly impact occupancy and reviews.

**THANK
YOU!**